Existing Database Structure

Simplified schema analysis with table purposes, columns, data types and key rules

Scope: This document is based on the supplied schema description. It lists the documented 27 tables, 10 enum types and the booking_meals view. Column notes retain important defaults, keys and constraints while omitting repetitive presentation detail.

1. Database overview

Object	Count	Role
Tables	27	Store application and operational data
Enum types	10	Standardise roles, statuses and categories
View	1	Provides meal-level booking data for reporting

2. Tables and columns

Other

1profiles

One row per portal user. Stores identity, role, and organisational assignment. Drives authorisation (role), scoping (hostel, unit) and sign-in (ldap_uid, password_hash).

Column	Type	Key / rule / notes
id	uuid	Required · Default: gen_random_uuid() · PK
email	text	Required · UNIQUE CHECK valid email shape
full_name	text	Required · CHECK length ≥ 2
role	user_role	Required
hostel_name	text	Nullable · FK → hostels(name) ON UPDATE CASCADE ON DELETE RESTRICT
department_or_club	text	Nullable
roll_number	text	Nullable
ldap_uid	text	Nullable · Unique index on lower(ldap_uid)
password_hash	text	Nullable
unit_id	uuid	Nullable · FK → units(id) ON DELETE RESTRICT
staff_category	text	Nullable · CHECK IN ('faculty','staff') or NULL

2guest_houses

The institute's guest houses (Bageshri, Hamsanandi). Drives guest-house selection on booking forms, room availability, and whether meals are offered.

Column	Type	Key / rule / notes
id	uuid	Required · Default: gen_random_uuid() · PK
name	text	Required · UNIQUE
total_rooms	integer	Required · Default: 0 · Maintained by trigger
serves_meals	boolean	Required · Default: false

3rooms

Physical rooms within a guest house. Used for the allocation grid, occupancy queries, and capacity calculations.

Column	Type	Key / rule / notes
id	uuid	Required · Default: gen_random_uuid() · PK
guest_house_id	uuid	Required · FK → guest_houses(id) ON DELETE CASCADE
room_number	text	Required · UNIQUE with guest_house_id
room_type	room_type	Required
is_active	boolean	Required · Default: true

4bookings

The central table. One row per booking request from submission through the approval pipeline to checkout and invoicing. Holds all request details: dates, guests, meals, debit head, and current status.

Column	Type	Key / rule / notes
id	uuid	Required · Default: gen_random_uuid() · PK
user_id	uuid	Required · FK → profiles(id)
guest_house_id	uuid	Required · FK → guest_houses(id)
user_role	user_role	Required · Role at time of booking
status	booking_status	Required · Current pipeline stage
purpose_of_visit	text	Required
check_in	timestamptz	Required
check_out	timestamptz	Required · CHECK check_out \> check_in
rooms_requested	integer	Required · CHECK 0–10; 0 for meals_only
booking_reference_id	text	Required · UNIQUE
rejection_reason	text	Nullable
alumni_id_url	text	Nullable · Upload path for alumni ID card
custom_fields	jsonb	Nullable · Snapshot of custom field answers
created_at	timestamptz	Required · Default: now()
updated_at	timestamptz	Required · Default: now() · Auto-updated by trigger
meals	jsonb	Required · Default: '\[\]' · Per-day meal plan array
has_infant	boolean	Required · Default: false
booking_type	booking_type	Required · Default: 'personal'
alumni_name	text	Nullable · Required when alumni booking
alumni_roll_number	text	Nullable · Required when alumni booking
service_type	service_type	Required · Default: 'room'
meal_preference	meal_preference	Nullable · Legacy single-answer preference
meal_guest_count	integer	Nullable · 1–100 for meals_only
pets_policy_acknowledged	boolean	Required · Default: false
pets_policy_acknowledged_at	timestamptz	Nullable
has_foreign_national	boolean	Required · Default: false
created_by	uuid	Nullable · FK → profiles(id) ON DELETE SET NULL
on_behalf_of_name	text	Nullable
on_behalf_of_email	text	Nullable
on_behalf_of_phone	text	Nullable
debit_head	text	Nullable · Budget code (project, personal_funds, etc.)
debit_details	text	Nullable · Project number+title
debit_document_url	text	Nullable · Approval document upload
office_approval	text	Nullable · CHECK IN ('direct','hod')
project_id	uuid	Nullable · FK → projects(id) ON DELETE SET NULL
extension_requested_until	timestamptz	Nullable
extension_reason	text	Nullable
extension_requested_at	timestamptz	Nullable
no_show_released_at	timestamptz	Nullable
privacy_notice_version	text	Nullable · Legacy — no longer written
privacy_consent_at	timestamptz	Nullable · Legacy — no longer written
search_text	tsvector	Required · Default: GENERATED · Full-text search vector
copy_to_emails	text\[\]	Nullable · Up to 25 CC addresses
debit_subhead	text	Nullable · CHECK only with debit_head \= 'project'
meal_diet_counts	jsonb	Nullable · {veg: N, non_veg: N}
fund_declaration_at	timestamptz	Nullable · When funds-approved was declared

5booking_rooms

One row per logical "room card" on a booking. A booking for 3 rooms has 3 rows. Guests are assigned to a room card. Legacy bookings have a single synthetic room.

Column	Type	Key / rule / notes
id	uuid	Required · Default: gen_random_uuid() · PK
booking_id	uuid	Required · FK → bookings(id) ON DELETE CASCADE
room_index	integer	Required · CHECK ≥ 1; UNIQUE with booking_id
room_type	room_type	Nullable
assigned_room_id	uuid	Nullable · FK → rooms(id) ON DELETE SET NULL
is_legacy	boolean	Required · Default: false

6booking_guests

One row per guest or infant on a booking, tied to a room card. Stores identity, relationship, ID documents, and citizenship. Occupancy trigger enforces per-room-card limits.

Column	Type	Key / rule / notes
id	uuid	Required · Default: gen_random_uuid() · PK
booking_id	uuid	Required · FK → bookings(id) ON DELETE CASCADE
name	text	Required
gender	guest_gender	Nullable
age	integer	Nullable · CHECK 0–120
relationship	text	Nullable
id_number	text	Nullable · Aadhaar or other ID
id_document_url	text	Nullable · Upload path
is_infant	boolean	Required · Default: false · Derived by trigger (age \< 5\)
booking_room_id	uuid	Nullable · FK → booking_rooms(id, booking_id)
citizenship	citizenship	Required · Default: 'indian'
nationality	text	Nullable · Required when other citizenship
passport_number	text	Nullable · Required when other citizenship

7booking_logs

Audit trail of every status change on a booking. Immutable append-only log. action_by_name is denormalised so history survives account deletion.

Column	Type	Key / rule / notes
id	uuid	Required · Default: gen_random_uuid() · PK
booking_id	uuid	Required · FK → bookings(id) ON DELETE CASCADE
action_by	uuid	Nullable · FK → profiles(id) ON DELETE SET NULL
action_by_name	text	Required · Denormalised actor name
previous_status	booking_status	Nullable · NULL on first entry
new_status	booking_status	Required
remarks	text	Nullable
timestamp	timestamptz	Required · Default: now()

8room_holds

One row per (booking, room) reservation. The guard column (stay \+ turnaround buffer) is checked by a GiST exclusion constraint to prevent double-booking at the DB level. Source of truth for which rooms are occupied.

Column	Type	Key / rule / notes
id	uuid	Required · Default: gen_random_uuid() · PK
booking_id	uuid	Required · FK → bookings(id) ON DELETE CASCADE
room_id	uuid	Required · FK → rooms(id) ON DELETE CASCADE
during	tstzrange	Required · Actual stay \[check_in, check_out)
created_at	timestamptz	Required · Default: now()
override_by	uuid	Nullable · FK → profiles(id) ON DELETE SET NULL
guard	tstzrange	Required · during \+ buffer; used by EXCLUDE

9app_settings

Key-value store for configurable rules: capacity limits, booking window, meal caps, approval routes, and the console password hash.

Column	Type	Key / rule / notes
key	text	Required · PK
value	jsonb	Required · Default: '{}'
updated_at	timestamptz	Required · Default: now() · Auto-updated by trigger

10hostels

Lookup table for hostel names (Malhar, Saveri, etc.). Profiles reference it so a warden's hostel scoping stays consistent.

Column	Type	Key / rule / notes
name	text	Required · PK CHECK length 1–60
created_at	timestamptz	Required · Default: now()

11units

Organisational tree: departments, clubs, councils, and offices. Drives HOD approval routing, Faculty Advisor assignments, and office debit head narrowing.

Column	Type	Key / rule / notes
id	uuid	Required · Default: gen_random_uuid() · PK
name	text	Required · UNIQUE
kind	text	Required · CHECK IN ('department','club','council','office')
parent_id	uuid	Nullable · FK → units(id) ON DELETE RESTRICT
head_id	uuid	Nullable · FK → profiles(id) ON DELETE SET NULL
acting_head_id	uuid	Nullable · FK → profiles(id) ON DELETE SET NULL
created_at	timestamptz	Required · Default: now()
office_class	text	Nullable · CHECK IN ('officer','department') or NULL
hod_unit_id	uuid	Nullable · FK → units(id); CHECK ≠ id
faculty_advisor_id	uuid	Nullable · FK → profiles(id) ON DELETE SET NULL
secretary_email	text	Nullable · Clubs/councils only; CHECK valid email

12form_configs

Per-role booking form configuration (field modes, allowed guest houses, relationship options, custom fields). Edited via the Form Builder console.

Column	Type	Key / rule / notes
role	user_role	Required · PK
config	jsonb	Required · RoleFormConfig object
updated_at	timestamptz	Required · Default: now()

13email_outbox

Queued email notifications. Actions enqueue messages here; the dispatch worker claims and sends them. Supports idempotency, threading, retries, and scheduled delivery.

Column	Type	Key / rule / notes
id	uuid	Required · Default: gen_random_uuid() · PK
booking_id	uuid	Nullable · FK → bookings(id) ON DELETE SET NULL
event_key	text	Required
idempotency_key	text	Required · UNIQUE
to_emails	text\[\]	Required
cc_emails	text\[\]	Required · Default: '{}'
subject	text	Required
body_html	text	Required
body_text	text	Required
thread_root	text	Nullable · Thread root Message-ID
is_thread_root	boolean	Required · Default: false
status	email_status	Required · Default: 'QUEUED'
attempts	integer	Required · Default: 0
last_error	text	Nullable
scheduled_for	timestamptz	Required · Default: now()
sent_at	timestamptz	Nullable
created_at	timestamptz	Required · Default: now()
updated_at	timestamptz	Required · Default: now() · Auto-updated by trigger
attachments	jsonb	Required · Default: '\[\]'

14mail_templates

Editable wording for each mail event. The manager or developer can enable/disable events, change subject/intro/outro, and add CC addresses without touching code.

Column	Type	Key / rule / notes
event_key	text	Required · PK
enabled	boolean	Required · Default: true
subject	text	Nullable
intro	text	Nullable
outro	text	Nullable
cc_emails	text\[\]	Required · Default: '{}'
updated_at	timestamptz	Required · Default: now()

15projects

Research projects that can be debited for a booking. A booking with debit_head \= 'project' optionally references one here.

Column	Type	Key / rule / notes
id	uuid	Required · Default: gen_random_uuid() · PK
project_number	text	Required · CHECK regex pattern; unique on lower()
title	text	Required
pi_name	text	Nullable · Principal Investigator
active	boolean	Required · Default: true
created_by	uuid	Nullable · FK → profiles(id) ON DELETE SET NULL
created_at	timestamptz	Required · Default: now()
updated_at	timestamptz	Required · Default: now()

16tariffs

Rate card: how much a room, extra bed, or meal costs, scoped by guest house, booking type, requester role, room type, and effective date. Protected by a trigger that refuses changes to rates already in force.

Column	Type	Key / rule / notes
id	uuid	Required · Default: gen_random_uuid() · PK
guest_house_id	uuid	Nullable · FK → guest_houses(id); NULL \= all
item	text	Required · CHECK IN ('room','extra_bed','breakfast','lunch','dinner')
room_type	text	Nullable · CHECK IN ('single','double_sharing')
booking_type	text	Nullable · CHECK IN ('official','personal','alumni')
requester_role	text	Nullable · Scoped by requester category
rate	numeric(10,2)	Required · CHECK ≥ 0
effective_from	date	Required
note	text	Nullable
created_by	uuid	Nullable · FK → profiles(id) ON DELETE SET NULL
created_at	timestamptz	Required · Default: now()

17invoice_counters

Sequential invoice numbering per financial year. The issue_invoice() function atomically increments the counter.

Column	Type	Key / rule / notes
fy	text	Required · PK CHECK YYYY-YY format
last_seq	integer	Required · CHECK ≥ 0

18invoices

Invoice lifecycle from draft to issued to paid or cancelled. The document column is a frozen snapshot so an issued invoice is never recomputed. Protected by an immutability trigger.

Column	Type	Key / rule / notes
id	uuid	Required · Default: gen_random_uuid() · PK
booking_id	uuid	Required · FK → bookings(id) ON DELETE RESTRICT
status	text	Required · Default: 'draft' · CHECK IN ('draft','issued','paid','cancelled')
invoice_number	text	Nullable · UNIQUE
fy	text	Nullable
seq	integer	Nullable · UNIQUE with fy
document	jsonb	Required · Default: '{}' · Frozen invoice snapshot
meal_counts	jsonb	Nullable
debit_head	text	Nullable · Denormalised
project_number	text	Nullable · Denormalised
subtotal_rooms	numeric(12,2)	Required · Default: 0
subtotal_dining	numeric(12,2)	Required · Default: 0
total	numeric(12,2)	Required · Default: 0
gst_percent	numeric(5,2)	Required · Default: 0
gst_amount	numeric(12,2)	Required · Default: 0
grand_total	numeric(12,2)	Required · Default: 0
payment_mode	text	Nullable · CHECK IN ('cash','upi','account_transfer')
payment_reference	text	Nullable
paid_at	timestamptz	Nullable
paid_by	uuid	Nullable · FK → profiles(id)
issued_at	timestamptz	Nullable
issued_by	uuid	Nullable · FK → profiles(id)
cancelled_at	timestamptz	Nullable
cancelled_by	uuid	Nullable · FK → profiles(id)
cancel_reason	text	Nullable
replaces_invoice_id	uuid	Nullable · FK → invoices(id)
created_by	uuid	Nullable · FK → profiles(id)
created_at	timestamptz	Required · Default: now()
updated_at	timestamptz	Required · Default: now()
extra_charges	jsonb	Required · Default: '\[\]' · Up to 20 line items

19room_blocks

Maintenance / out-of-service periods for individual rooms. Prevents allocation during the blocked period via a GiST exclusion constraint.

Column	Type	Key / rule / notes
id	uuid	Required · Default: gen_random_uuid() · PK
room_id	uuid	Required · FK → rooms(id) ON DELETE CASCADE
during	tstzrange	Required · CHECK non-empty and bounded
reason	text	Required · CHECK length 3–300
created_by	uuid	Nullable · FK → profiles(id) ON DELETE SET NULL
created_at	timestamptz	Required · Default: now()

20sessions

Server-side sessions. The cookie carries a random token; only the SHA-256 hash is stored. Supports idle timeout (30 min), absolute expiry (12 h), revocation, token rotation, and MFA step-up.

Column	Type	Key / rule / notes
id	uuid	Required · Default: gen_random_uuid() · PK
user_id	uuid	Required · FK → profiles(id) ON DELETE CASCADE
token_hash	text	Required · UNIQUE
created_at	timestamptz	Required · Default: now()
last_seen_at	timestamptz	Required · Default: now()
idle_expires_at	timestamptz	Required · 30 min from last activity
absolute_expires_at	timestamptz	Required · 12 h from sign-in
revoked_at	timestamptz	Nullable
verified_at	timestamptz	Nullable · MFA step-up window
rotated_from	uuid	Nullable · FK → sessions(id)
ip	text	Nullable
user_agent	text	Nullable

21user_mfa

TOTP second-factor enrollment per developer account. Stores the encrypted secret, recovery codes, and confirmation state.

Column	Type	Key / rule / notes
user_id	uuid	Required · PK FK → profiles(id) ON DELETE CASCADE
secret_enc	text	Required · Encrypted TOTP secret
key_version	integer	Required · Default: 1
recovery_codes	jsonb	Required · Default: '\[\]'
last_step	bigint	Nullable · Replay protection
confirmed_at	timestamptz	Nullable
created_at	timestamptz	Required · Default: now()
updated_at	timestamptz	Required · Default: now()

22rate_limits

Sliding-window rate limiting stored in the database so it survives restarts and is shared across instances. Used for sign-in, console unlock, and OAuth throttling.

Column	Type	Key / rule / notes
key	text	Required · PK
window_start	timestamptz	Required · Default: now()
count	integer	Required · Default: 0

23privacy_requests

DPDP Act data-subject requests \- "Download my data" and "Ask for erasure" from My Bookings. The office handles them through Console → Security.

Column	Type	Key / rule / notes
id	uuid	Required · Default: gen_random_uuid() · PK
user_id	uuid	Required · FK → profiles(id) ON DELETE CASCADE
kind	text	Required · CHECK IN ('export','deletion')
status	text	Required · Default: 'open' · CHECK IN ('open','done','refused')
note	text	Nullable
response	text	Nullable
created_at	timestamptz	Required · Default: now()
handled_at	timestamptz	Nullable
handled_by	uuid	Nullable · FK → profiles(id)

24security_audit

Append-only security event log. Every Setting change, role change, unit edit, import, and sign-in is recorded. Protected by triggers that refuse UPDATE, DELETE (except via purge after 180 days), and TRUNCATE.

Column	Type	Key / rule / notes
id	bigint	Required · Default: GENERATED ALWAYS AS IDENTITY · PK
at	timestamptz	Required · Default: now()
actor_id	uuid	Nullable · FK → profiles(id) ON DELETE SET NULL
actor_name	text	Required · Denormalised
actor_role	text	Nullable
event	text	Required
target	text	Nullable
details	jsonb	Required · Default: '{}'
ip	text	Nullable
user_agent	text	Nullable

25academic_records

Institute academic records pasted in by the office. Used to auto-fill and lock parents' names on student booking forms, and to show the Requester Details card.

Column	Type	Key / rule / notes
id	uuid	Required · Default: gen_random_uuid() · PK
kind	academic_record_kind	Required · UNIQUE with lower(email)
email	text	Required
name	text	Nullable
department	text	Nullable
phone	text	Nullable
roll_number	text	Nullable
program	text	Nullable
father_name	text	Nullable
mother_name	text	Nullable
guardian_name	text	Nullable
hostel	text	Nullable
employee_id	text	Nullable
employee_type	text	Nullable
office_number	text	Nullable
head_name	text	Nullable
head_email	text	Nullable
representative_type	text	Nullable
faculty_in_charge_email	text	Nullable
imported_by	uuid	Nullable · FK → profiles(id) ON DELETE SET NULL
created_at	timestamptz	Required · Default: now()
updated_at	timestamptz	Required · Default: now()

26user_import_batches

Undo record for spreadsheet imports of user accounts. Stores accounts created and previous values of accounts changed, so "Undo" can restore the old state.

Column	Type	Key / rule / notes
id	uuid	Required · Default: gen_random_uuid() · PK
at	timestamptz	Required · Default: now()
actor_id	uuid	Nullable · FK → profiles(id) ON DELETE SET NULL
actor_name	text	Required
status	text	Required · Default: 'applying' · CHECK IN ('applying','applied','failed','reverted')
columns_used	text	Required · Default: ''
added	jsonb	Required · Default: '\[\]'
updated	jsonb	Required · Default: '\[\]'
unchanged_count	integer	Required · Default: 0 · CHECK ≥ 0
error	text	Nullable
applied_at	timestamptz	Nullable
reverted_at	timestamptz	Nullable
reverted_by	uuid	Nullable · FK → profiles(id)
reverted_actor_name	text	Nullable

27official_email_whitelist

Email addresses allowed to submit official bookings. Managed in Console → Settings. Checked by isWhitelistedOfficial().

Column	Type	Key / rule / notes
email	text	Required · PK CHECK valid lower-cased email
created_at	timestamptz	Required · Default: now()

3. Enum types

Enum types limit a column to a defined set of values. Values below are shown separately for readability; names and values follow the supplied schema source.

user_role

```
student • employee • official • club • warden • faculty • advisor • iar_cell • gh_manager • developer • iar_student_cell • gh_caretaker
```

booking_status

```
PENDING_WARDEN_REVIEW • PENDING_FA_REVIEW • PENDING_IAR_REVIEW • PENDING_GH_MANAGER • APPROVED • REJECTED • CANCELLED • CANCELLATION_REQUESTED • CANCELLATION_APPROVED • OCCUPIED • VACATED • PENDING_HOD • MISSED
```

room_type

```
single • double_sharing
```

guest_gender

```
male • female • other
```

booking_type

```
official • personal • alumni
```

email_status

```
QUEUED • SENDING • SENT • FAILED
```

citizenship

```
indian • other
```

service_type

```
room • room_meals • meals_only
```

meal_preference

```
veg • non_veg
```

academic_record_kind

```
student • employee • office • student_rep • alumni_office • warden
```

4. View

booking_meals — A flattened view with one row per booking, meal date and meal type. It derives data from bookings.meals and exposes booking_id, guest_house_id, status, meal_preference, meal_date, meal_type and guest_count.

5. How to read the column notes

Required / Nullable indicates the documented nullability. **Default** shows a documented default value. **PK** identifies a primary key. **FK** identifies a foreign-key relationship. Other notes summarise documented checks, uniqueness rules, generated values or business meaning.

6. Analysis boundary

This is a structural reference derived from the supplied Markdown file. It does not verify the live database, actual row counts, indexes not listed in the source, query performance, trigger implementation, permissions or data quality. Those require direct inspection of the PostgreSQL database.